

Solution

1. Weighted Mean:

$$\text{Weighted Mean} = \frac{25 \times 58 + 29 \times 62}{25 + 29} = \frac{3248}{54} = 64.15$$

2. The mode is valuable in marketing because it highlights the most frequent and popular choice, helping businesses understand customer preferences and design strategies that align with the majority.

- Mode Represents the Most Frequent Preference
- Understanding Popularity
- When Data is Skewed
- Simple and Effective

3.

Time	frequency (f)	Midpoint (x_i)	$f_i \times x_i$	c.f	
0-10	14	5	70	14	
10-20	26	15	390	40	
20-30	64	25	1600	104	Median class, Mode Class
30-40	46	35	1610	150	
40-50	28	45	1260	178	
50-60	16	55	880	194	
60-70	8	65	520	202	
70-80	4	75	300	206	
	$\Sigma f_i = 206$		$\Sigma(f_i \cdot x_i) = 6630$		

$$\text{Mean} = \frac{\Sigma(f_i \cdot x_i)}{\Sigma f_i} = \frac{6630}{206} = 32.18$$

$$\text{Median} = l + \frac{h \left(\frac{n}{2} - c.f \right)}{f} = 20 + \frac{10 \left(\frac{206}{2} - 40 \right)}{64} = 29.84$$

$$\text{Mode} = l + h \times \frac{(f_m - f_1)}{2f_m - f_1 - f_2} = 20 + 10 \times \frac{64 - 26}{2(64) - 26 - 46} = 26.76$$

- As an employer, I would focus on the **mean** commute time, as it represents the overall average commute time for all employees.
- As a trade union representative, I would focus on the **median** commute time. The median is more representative of the central tendency in a situation where there might be a few employees with extremely long commute times that could skew the mean.

4. Given that a and b are positive numbers.

$$AM = \frac{a + b}{2}, \quad GM = (a \times b)^{\frac{1}{2}}, \quad HM = \frac{2ab}{a + b}$$

$$HM = \frac{ab}{\frac{a+b}{2}} = \frac{GM^2}{AM}$$

AM, GM and HM between numbers:

$$AM = \frac{45 + 21 + 77 + 68}{4} = 52.75$$

$$GM = (45 \times 21 \times 77 \times 68)^{\frac{1}{4}} = 47.16$$

$$HM = \frac{4}{\left(\frac{1}{45} + \frac{1}{21} + \frac{1}{77} + \frac{1}{68}\right)} = 41.01$$